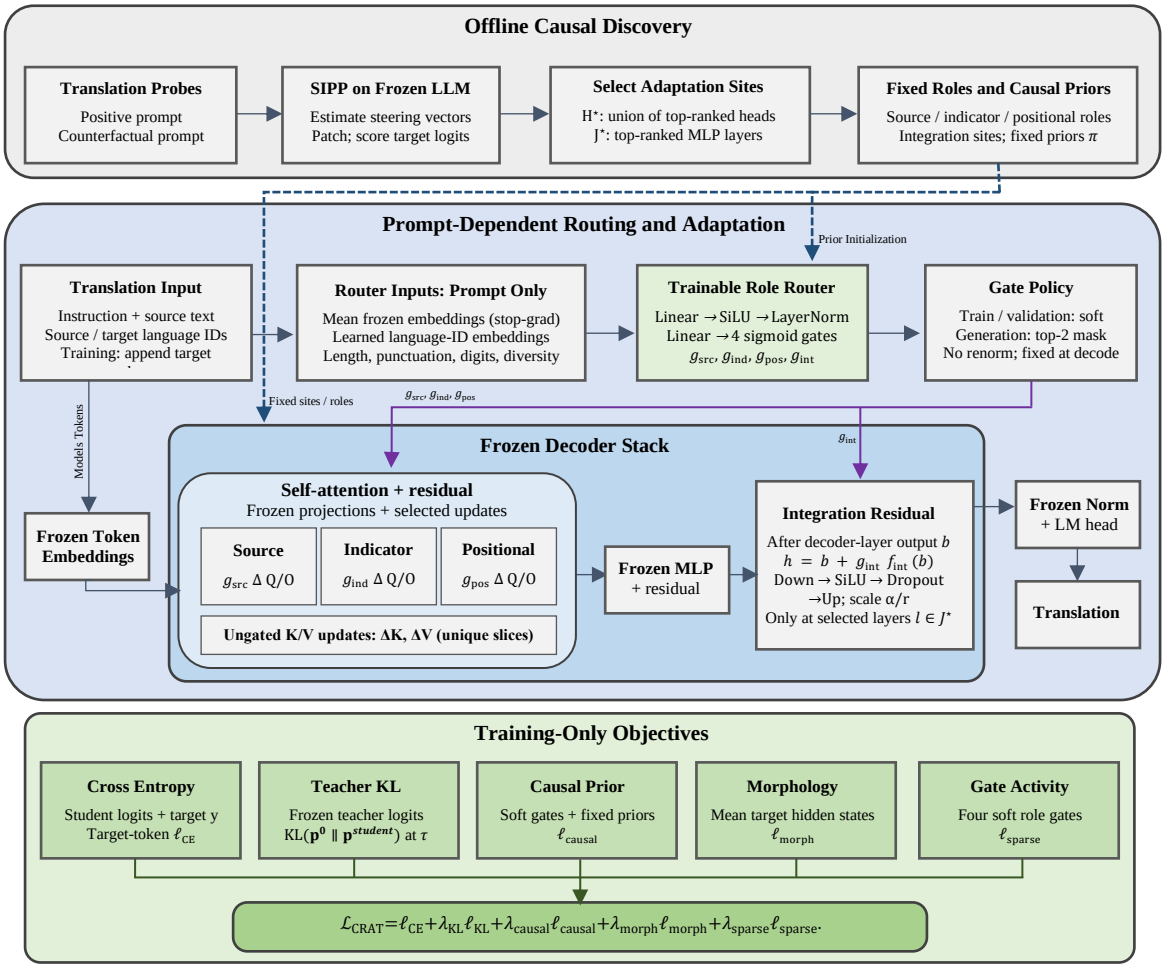

Offline Causal Discovery
Translation Probes
Positive prompt
Counterfactual prompt
SIPP on Frozen LLM
Estimate steering vectors
Patch; score target logits
Select Adaptation Sites
H*: union of top-ranked heads
J*: top-ranked MLP layers
Fixed Roles and Causal Priors
Source / indicator / positional roles
Integration sites; fixed priors π
Prompt-Dependent Routing and Adaptation
Prior Initialization
Translation Input
Instruction + source text
Source / target language IDs
Training: append target
Router Inputs: Prompt Only
Mean frozen embeddings (stop-grad)
Learned language-ID embeddings
Length, punctuation, digits, diversity
Trainable Role Router
Linear → SiLU → LayerNorm
Linear → 4 sigmoid gates
g_src, g_ind, g_pos, g_int
Gate Policy
Train / validation: soft
Generation: top-2 mask
No renorm; fixed at decode
Models Tokens
Fixed sites / roles
g_src, g_ind, g_pos
g_int
Frozen Token Embeddings
Frozen Decoder Stack
Self-attention + residual
Frozen projections + selected updates
Source
g_src Δ Q/O
Indicator
g_ind Δ Q/O
Positional
g_pos Δ Q/O
Ungated K/V updates: ΔK, ΔV (unique slices)
Frozen MLP
+ residual
Integration Residual
After decoder-layer output b
h = b + g_int f_int (b)
Down → SiLU → Dropout
→Up; scale α/r
Only at selected layers l ∈ J*
Frozen Norm
+ LM head
Translation
Training-Only Objectives
Cross Entropy
Student logits + target y
Target-token ℓ_CE
Teacher KL
Frozen teacher logits
KL(p^0 ‖ p^student) at τ
Causal Prior
Soft gates + fixed priors
ℓ_causal
Morphology
Mean target hidden states
ℓ_morph
Gate Activity
Four soft role gates
ℓ_sparse
L_CRAT = ℓ_CE + λ_KL ℓ_KL + λ_causal ℓ_causal + λ_morph ℓ_morph + λ_sparse ℓ_sparse.